

5. Rational Function Fields.^[316:1]

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The following questions originally arose from conversations with E. Fischer. Some of them, moreover, had already been raised and settled by E. Steinitz in the special case of one variable — and under more general assumptions on the coefficient domain.^[316:2]

By a “rational function field” I mean a field whose elements are rational functions in n variables with coefficients in a prescribed field of scalars, which may in particular contain all complex numbers. Examples are the field of symmetric functions in n quantities, or, more generally, the totality of rational functions in n variables that admit the permutations of a given group (the *Lagrange genus domains*, that is, the fields attached to such permutation groups); and the field of invariants.

There is a characteristic difference between the first two fields just mentioned and the last: the former contain n algebraically independent functions, the elementary symmetric functions, whereas the number of algebraically independent invariants is always smaller than the number of variables. It is therefore important that fields of the second kind can in general be associated bijectively with fields of the first kind by replacing some of the variables with numbers. In what follows I shall therefore restrict myself to fields of the first kind, that is, fields with n algebraically independent functions; by virtue of this association, the results then hold in general.

The questions group themselves around three basic concepts: rational basis, minimal basis, and integrality basis.

1. By a “rational basis” I mean a finite collection of functions of the field such that every function of the field can be expressed rationally in terms of this finite collection, with coefficients from the prescribed coefficient domain. Simple arguments — which, in the case of one variable, are also found in E. Steinitz, loc. cit. — prove the existence of a rational basis for every rational function field. For example, by Lagrange’s theorem, the elementary symmetric functions together with a function belonging to the group form a rational basis for the Lagrange genus domains mentioned above.

2. Every rational basis must contain at least n functions (for fields of the first kind); if it contains exactly n functions, which must then be algebraically independent, I call it a “minimal basis.” The question of the existence of a minimal basis can be answered in part by theorems of Lüroth, Castelnuovo, and Enriques.^[317:1] It follows that a minimal basis always exists for fields in one variable: the Lüroth function of the field, uniquely determined up to a linear fractional transformation (cf. Steinitz § 24). For fields in two variables, a minimal basis exists whenever, and in general only when, an algebraic extension of the coefficient domain is allowed; for three or more variables, by contrast, a minimal basis does not in general exist at all. No attempt has yet been made to characterize the special fields that possess a minimal basis.

I have pursued further the question of the minimal basis for Lagrange genus domains. Here it takes on the following meaning: “If the Lagrange genus domain belonging to the group G possesses a minimal basis, then the totality of equations with prescribed group G can be constructed rationally by a parametric representation; this can be done over every field of scalars containing the coefficient domain of the minimal basis — hence, in particular, over any field of scalars whatever if that coefficient domain is the field of rational numbers.”

^[316:1]Lecture delivered at the Meeting of Natural Scientists in Vienna, 1913.

^[316:2]E. Steinitz: *Algebraische Theorie der Körper* (especially § 24). Crelle’s Journal 137. 1910.

^[317:1]J. Lüroth: *Beweis eines Satzes über rationale Kurven*. Math. Ann. 9. 1875. — G. Castelnuovo: *Sulla razionalità delle involuzioni piane*. Math. Ann. 44. 1893. — F. Enriques: *Sopra una involuzione non razionale dello spazio*. Rend. Acc. Linc. vol. 21. 21 Jan. 1912.

The simplest example is provided by the symmetric group: here the elementary symmetric functions form a minimal basis with rational coefficients. The corresponding parametric representation is simply the equation with indeterminate coefficients. Hilbert's irreducibility theorem shows how one passes from this to arbitrarily many affect-free equations over every field of scalars.

The existence of a minimal basis for all groups occurring in equations of degrees 3 and 4 now follows directly from the theorems of Lüroth and Castelnuovo; moreover, this minimal basis has rational coefficients. Thus, over any field of scalars, the totality of equations of degrees 3 and 4 with prescribed group can be constructed rationally. For cyclic and dihedral groups there is a minimal basis whose coefficient domain is the field of the n th roots of unity; the same holds for several further metacyclic groups. The question whether the existence of a minimal basis characterizes certain groups must remain open.

3. To reach the final basis concept, I consider the totality of polynomials contained in the field. By an "integrality basis" I mean a finite collection of these polynomials such that every polynomial in the field can be expressed as a polynomial expression in this finite collection, with coefficients from the prescribed coefficient domain. The question of the existence of an integrality basis — which includes, for example, the finiteness of the system of invariants as a special case — was posed by Hilbert in his "Mathematical Problems"^[318:1] in a somewhat different form, as the problem of relatively integral functions.

For the moment I can give only one class of fields for which an integrality basis exists. Namely, if the field contains a system of n polynomials such that the homogeneous resultant of the highest-degree homogeneous terms of these polynomials is nonzero, then an integrality basis exists; it is furnished by the coefficients, with respect to all the u 's, of the equation for the linear form that is irreducible over the field:

$$u_0 = u_1x_1 + u_2x_2 + \cdots + u_nx_n. \quad [318:2]$$

If, in particular, the value of the resultant is a unit of the coefficient domain, then the integrality of the representation is also assured. The Lagrange genus domains provide an example of this class of fields, since the resultant of the elementary symmetric functions has the value ± 1 . A further example (without integrality) is given by all fields containing only one algebraically independent polynomial; here the integrality basis is identical with the Lüroth function of the smallest subfield containing all polynomials. Thus, as is well known, all polynomial projective invariants of a quadratic form in n variables are polynomial functions of the discriminant.

The stated condition is sufficient but by no means necessary for the existence of an integrality basis. One can easily construct fields in which the resultant of any n polynomials vanishes and which nevertheless possess an integrality basis furnished by the coefficients of that irreducible equation. The question arises whether, even in the most general case, the coefficients of that equation furnish the integrality basis.

^[318:1]D. Hilbert: *Mathematische Probleme*. Lecture, Paris 1900. Problem 14. Göttinger Nachr. 1900.

^[318:2]For one variable, this irreducible equation occurs in E. Steinitz's proof of Lüroth's theorem.